

Seat No.: _____

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY
M.Sc. Digital Forensics and Information Security - Semester - I - December-2025

Subject Code: CTMSDFIS24 SI PI
Subject Name: Computer Forensics
Time: 2.00 PM TO 5.00 PM
Total Marks: 100

Date: 15/12/2025

Instructions:

1. Write down each question on separate page.
2. Attempt all questions.
3. Make suitable assumptions wherever necessary.
4. Figures to the right indicate full marks.

		Marks
Q.1	Attempt any three.	
(a)	Explain using examples, what do you understand by little endian and big endian number formats. Which format is usually used by computers and why?	08
(b)	What are secondary storage devices? Give an example. Give the full forms and describe in a line or two: MBR, GPT, and VBR.	08
(c)	The role of the Operating System is crucial in functioning of a computer system. Explain the following roles of the Operating System: i) Process Management ii) Memory Management	08
(d)	What do you understand by booting? What are the steps in the boot process? Can malware impact boot process?	08
Q.2	Attempt any three.	
(a)	What do you understand by digital forensics? What are the steps in a digital forensic investigation. Mention any two branches of digital forensics.	08
(b)	What is Locard's Principle of Exchange? An evidence should be relevant to the case of investigation. Mention any other two properties of evidence. Write any four precautions that must be taken while labelling, packaging, and transportation of digital evidence.	08
(c)	What do you understand by disk acquisition? What is the role of write-blockers during acquisition process. What is a chain of custody? What is hashing and why is hashing important in forensics?	08
(d)	How many forensic copies of the evidence should an analyst make? Name any two tools that can be used to analyse disk images. What is data carving? What is the importance of data recovery in forensic investigations?	08

Q.3

Attempt any three.

- (a) What is a filesystem? What is the importance of a file system? Give any two examples of File Systems. Some filesystems maintain journaling. What do you understand by the term 'Journaling'? 08
- (b) What is \$MFT in Windows NTFS file system? Give any one feature of the NTFS file system. From a forensic point of view, explain what happens when a Windows file is deleted? What is slack space? 08
- (c) What is Windows registry? What is the importance of HKLM (Hives Key Local Machine)? What is the forensic significance of NTUSER.DAT? Name any two tools used for analysis of registry artefacts. 08
- (d) How can logs be helpful in forensic analysis? With reference to Windows Event Logs explain what are Application logs, Security logs, and System Logs. 08

Q.4

Attempt any two.

- (a) Name and explain the timestamps maintained by Windows. What is timestamping? How is it detected? 07
- (b) With reference to the Linux operating system, what is the difference between kernel space and user space? Which (kernel space or user space) is targeted by malware and why? What is a shell in Linux? 07
- (c) What are system calls? Give examples of any two system calls. What is the significance of the following directories: 07
 - i) /boot
 - ii) /lib
 - iii) /var
 - iv) /usr

Q.5

Attempt any two.

- (a) What is EXT file system? What are the following in EXT file system: superblocks, inodes, block groups, data blocks, and bitmaps. 07
- (b) Write a short note on the structure of the APFS (Apple Proprietary File System). 07
- (c) What are user artefacts? Name any four Mac Artifacts. What is the similarity. What do you understand by the following mechanisms: 07
 - i) Atomic Safe-Save
 - ii) Copy-on Write

END OF PAPER